

Q1 Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

A food delivery application calculates the final bill for a customer's order by considering the food cost, delivery charges, and coupon discounts.

Create two interfaces:

DeliveryCharge with a method calculateDeliveryCharge().

CouponDiscount with a method calculateDiscount().

Create a class FoodOrder that implements both interfaces.

The delivery charge is calculated as follows:

If the food cost is ₹500 or more, the delivery charge is ₹0.

Otherwise, the delivery charge is ₹50.

The coupon discount is calculated as follows:

If the coupon code is SAVE10, the customer receives a 10% discount on the food cost.

Otherwise, no discount is applied.

The final bill is calculated using the formula:

Final Bill = Food Cost + Delivery Charge - Discount

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

Coupon Code

Output Format

Food Cost : <food_cost>

Delivery Charge : <delivery_charge>

Discount : <discount>

Final Bill : <final_bill>

Sample Input

600

SAVE10

Sample Output

Food Cost : 600.0

Delivery Charge : 0.0

Discount : 60.0

Final Bill : 540.0

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5
6         double Foodcost = sc.nextDouble();
7         sc.nextLine();
8         String Coupon = sc.nextLine();
9
10        Foodorder f = new Foodorder(Foodcost, Coupon);
11        f.displayDetails();
12    }
13 }
14 interface DeliveryCharge {
15     void calculateDeliveryCharge();
16 }
17 interface CouponDiscount {
18     void calculateDiscount();
19 }
20 class Foodorder implements DeliveryCharge, CouponDiscount {
21     double Foodcost;
22     String Coupon;
23     double DeliveryCharge;
24     double Discount;
25     double FinalBill;
26
27     Foodorder(double Foodcost, String Coupon) {
28         this.Foodcost = Foodcost;
29         this.Coupon = Coupon;
30     }
31     public void calculateDeliveryCharge() {
32         if (Foodcost >= 500)
33             DeliveryCharge = 0;
34         else
35             DeliveryCharge = 50;
36     }
37     public void calculateDiscount() {
38         if (Coupon.equals("SAVE10"))
```

```
23         double DeliveryCharge;
24         double Discount;
25         double FinalBill;
26
27         Foodorder f = new Foodorder(Foodcost, Coupon) {
28             this.Foodcost = Foodcost;
29             this.Coupon = Coupon;
30         }
31         public void calculateDeliveryCharge() {
32             if (Foodcost >= 500)
33                 DeliveryCharge = 0;
34             else
35                 DeliveryCharge = 50;
36         }
37         public void calculateDiscount() {
38             if (Coupon.equals("SAVE10"))
39                 Discount = Foodcost * 0.10;
40             else
41                 Discount = 0;
42         }
43         FinalBill = Foodcost + DeliveryCharge - Discount;
44     }
45     void displayDetails() {
46         calculateDeliveryCharge();
47         calculateDiscount();
48
49         System.out.println("Food Cost : " + Foodcost);
50         System.out.println("Delivery Charge : " + DeliveryCharge);
51         System.out.println("Discount : " + Discount);
52         System.out.println("Final Bill : " + FinalBill);
53     }
54 }
```

Q2 Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Problem Statement

A hotel wants to generate the final bill for its guests by adding a service charge to the room rent.

Create a class Room that stores the following details:

Room Number

Guest Name

Room Rent

Create an interface ServiceCharge with a method:

double calculateServiceCharge();

Create a class HotelBill that extends the Room class and implements the ServiceCharge interface.

The service charge is calculated as follows:

If the room rent is ₹5000 or more, the service charge is 12% of the room rent.

Otherwise, the service charge is 8% of the room rent.

The final bill is calculated as:

Final Bill = Room Rent + Service Charge

Display the room details, room rent, service charge, and final bill.

Input Format

Room Number

Guest Name

Room Rent

Output Format

Room Number : <room_number>

Guest Name : <guest_name>

Room Rent : <room_rent>

Service Charge : <service_charge>

Final Bill : <final_bill>

Sample Input

205

Ananya

6000

Sample Output

Room Number : 205

Guest Name : Ananya

Room Rent : 6000.0

Service Charge : 720.0

Final Bill : 6720.0

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s1=new Scanner(System.in);
5         int roomNumber=s1.nextInt();
6         s1.nextLine();
7         String guestName=s1.nextLine();
8         double roomRent=s1.nextDouble();
9
10        HotelBill h=new HotelBill(roomNumber,guestName,roomRent);
11        h.calculateServiceCharge();
12        h.displayDetails();
13    }
14 }
15 interface ServiceCharge{
16     void calculateServiceCharge();
17 }
18 class Room{
19     int roomNumber;
20     String guestName;
21     double roomRent;
22
23     Room(int roomNumber,String guestName,double roomRent){
24         this.roomNumber=roomNumber;
25         this.guestName=guestName;
26         this.roomRent=roomRent;
27     }
28 }
29 class HotelBill extends Room implements ServiceCharge{
30     double serviceCharge;
31     double finalBill;
32
33     HotelBill(int roomNumber,String guestName,double roomRent){
34         super(roomNumber,guestName,roomRent);
35     }
```

Input

205
Ananya
6000

Output

Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0

Visible Test Cases

Test Case 1 ☒

Test Case 2 ☒

Test Case 3 ☒

10.00 / 10 marks

```
36     public void calculateServiceCharge(){
37         if(roomRent>=5000)
38             serviceCharge=roomRent*0.12;
39         else
40             serviceCharge=roomRent*0.08;
41
42         finalBill=roomRent+serviceCharge;
43     }
44     void displayDetails(){
45         System.out.println("Room Number : "+roomNumber);
46         System.out.println("Guest Name : "+guestName);
47         System.out.println("Room Rent : "+roomRent);
48         System.out.println("Service Charge : "+serviceCharge);
49         System.out.println("Final Bill : "+finalBill);
50     }
51 }
```